

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Bachelor of Science in Data Science

Modern Application Development I (CS2003)



PROJECT REPORT

Musafir Treks

A Role-Based Trekking Management Web Application

Submitted By

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1. Student Details

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About Me

I am an undergraduate student pursuing a Bachelor of Technology in Electronics and Communication Engineering with a specialization in Artificial Intelligence and Machine Learning, from MIT-WPU Kothrud, Pune while simultaneously pursuing the Bachelor of Science in Data Science at the Indian Institute of Technology Madras. My areas of interest include Artificial Intelligence, Machine Learning, Computer Vision, and Full Stack Web Development. I enjoy designing practical software solutions that combine intelligent decision-making with user-friendly interfaces.

Through this project, I explored the integration of role-based web applications, database management, analytics, and AI-assisted recommendations into a unified trekking management platform.

2. Project Details

Musafir Treks – A Role-Based Trekking Management Web Application.

Problem Statement

Trekking organizations often rely on manual methods to manage trek registrations, participant records, guide assignments, and booking information. Such approaches are prone to errors including duplicate bookings, overbooking, inconsistent participant tracking, and inefficient communication between administrators, trek guides, and trekkers. There is a need for a centralized web-based system that simplifies trek management while providing users with a seamless booking experience.

Approach

Musafir Treks was developed as a role-based web application using the Flask framework following a modular architecture. The application supports three different user roles—Administrator, Trek Staff, and Trekker—each with dedicated dashboards and functionalities. SQLAlchemy ORM was used to manage the SQLite database efficiently while Flask-Login provides secure authentication and session management. The application integrates the OpenWeather API to display live weather information for trekking destinations and includes an AI-assisted recommendation module that suggests treks based on user preferences. Interactive dashboards powered by Chart.js provide analytical insights for administrators and trek staff, resulting in an intuitive and scalable trekking management solution.

3. AI / LLM Declaration

ChatGPT (GPT-5.5) was used during the development of this project as a coding assistant for debugging, UI refinement, documentation support, code refactoring, and feature brainstorming. AI assistance was limited to suggesting improvements, identifying bugs, and helping with project documentation.

The complete application architecture, database design, implementation, testing, integration, and verification of all functionalities were carried out manually by the author.

The approximate AI assistance in this project is **15–20%**, primarily for development guidance and documentation.

4. Technologies and Frameworks Used

Technology/Library	Purpose
Python	Primary programming language
Flask	Backend web framework
SQLAlchemy	ORM for database operations
SQLite	Local relational database
Flask-Login	User authentication and session management
Werkzeug Security	Password hashing and verification
HTML5	Web page structure
CSS3	User interface styling
Bootstrap	Responsive frontend components
Jinja2	Dynamic template rendering
JavaScript	Client-side interactivity
Chart.js	Dashboard analytics and visualization
Requests	OpenWeather API integration
Git & GitHub	Version control and source code management

5. Database Schema / ER Diagram

User

Stores all registered users, administrators, and trek staff.

Attributes

- id (Primary Key)
 - name
 - email
 - password
 - role
 - status
-

Trek

Stores complete information about trekking expeditions.

Attributes

- id (Primary Key)
- name
- location
- difficulty
- duration
- available_slots
- staff_id (Foreign Key)

- status
 - start_date
 - end_date
 - description
 - image
 - created_at
 - price
-

Booking

Stores trek booking information made by users.

Attributes

- id (Primary Key)
 - user_id (Foreign Key)
 - trek_id (Foreign Key)
 - booking_date
 - status
 - payment_status
-

Staff Profile

Stores additional information related to trek staff.

Attributes

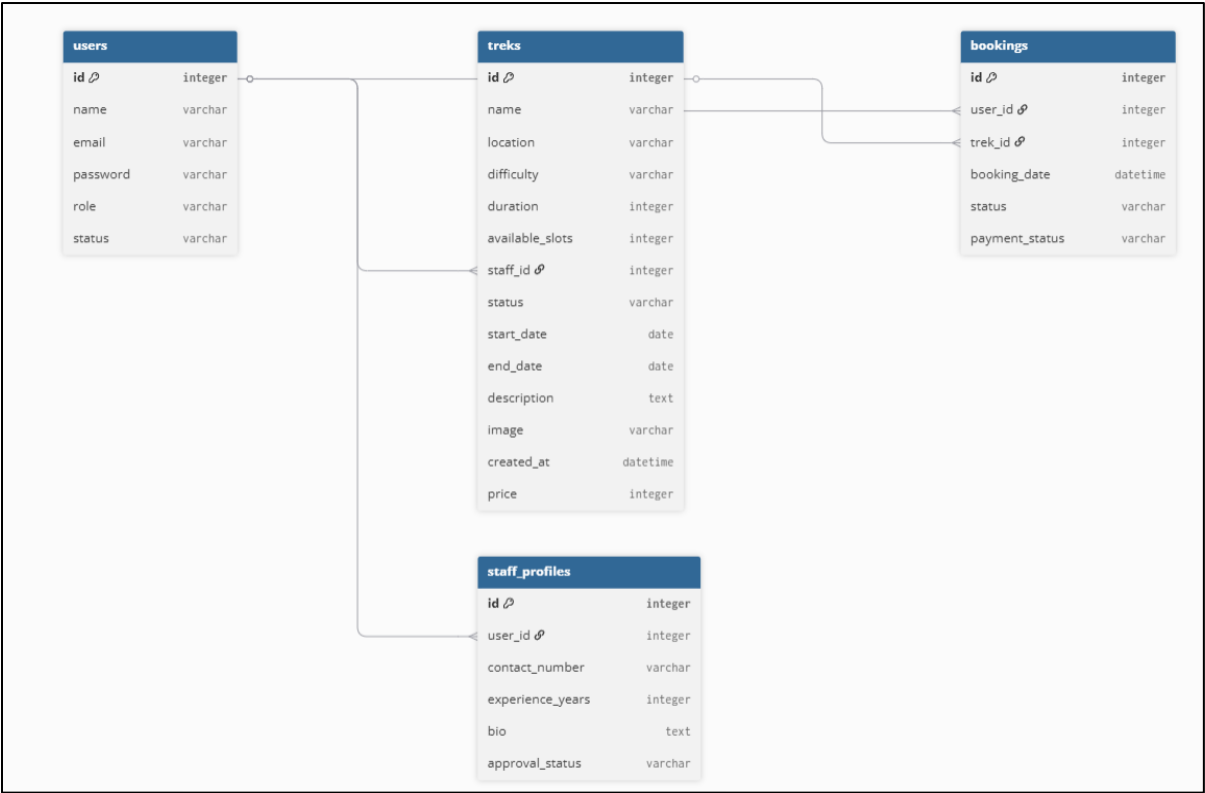
- id (Primary Key)

- user_id (Foreign Key)
 - contact_number
 - experience_years
 - bio
 - approval_status
-

Relationships

- One User can make multiple Bookings.
- One Trek can have multiple Bookings.
- One Trek Staff (User) can manage multiple Treks.
- Each Trek is assigned to one Trek Staff member.
- Each Staff Profile belongs to exactly one Staff member.
- Every Booking belongs to exactly one User and one Trek.

ER Diagrams



6. API Resource Endpoints

Endpoint	Method	Description
/	GET, POST	User Login
/register	GET, POST	User Registration
/forgot-password	GET, POST	Password Reset
/user	GET	User Dashboard
/admin	GET	Administrator Dashboard
/staff	GET	Trek Staff Dashboard
/recommend	GET, POST	AI Trek Recommendation
/trek/<id>	GET	Trek Details
/book/<id>	GET	Book Trek
/cancel-booking/<id>	GET	Cancel Trek Booking

YAML API Definition

The application uses Flask route-based endpoints. Hence, a separate OpenAPI/YAML specification was not implemented.

7. Architecture and Features

Architecture Overview

The project follows a modular Flask architecture comprising the following components:

- **app.py** – Main application entry point containing all routes and controller logic.
 - **models.py** – SQLAlchemy database models defining application entities and relationships.
 - **extensions.py** – Database and authentication initialization.
 - **templates/** – Jinja2 templates used for rendering dynamic web pages.
 - **static/** – CSS stylesheets, images, JavaScript files, and application assets.
 - **instance/** – SQLite database used for persistent data storage.
-

Implemented Features

User Module

- User registration and secure login
- Password hashing using Werkzeug
- Browse and search treks
- AI-based trek recommendations
- Weather information for trekking locations
- Trek booking

- Booking cancellation
- Booking history
- Trek status tracking

Screenshots of User-Dashboard:

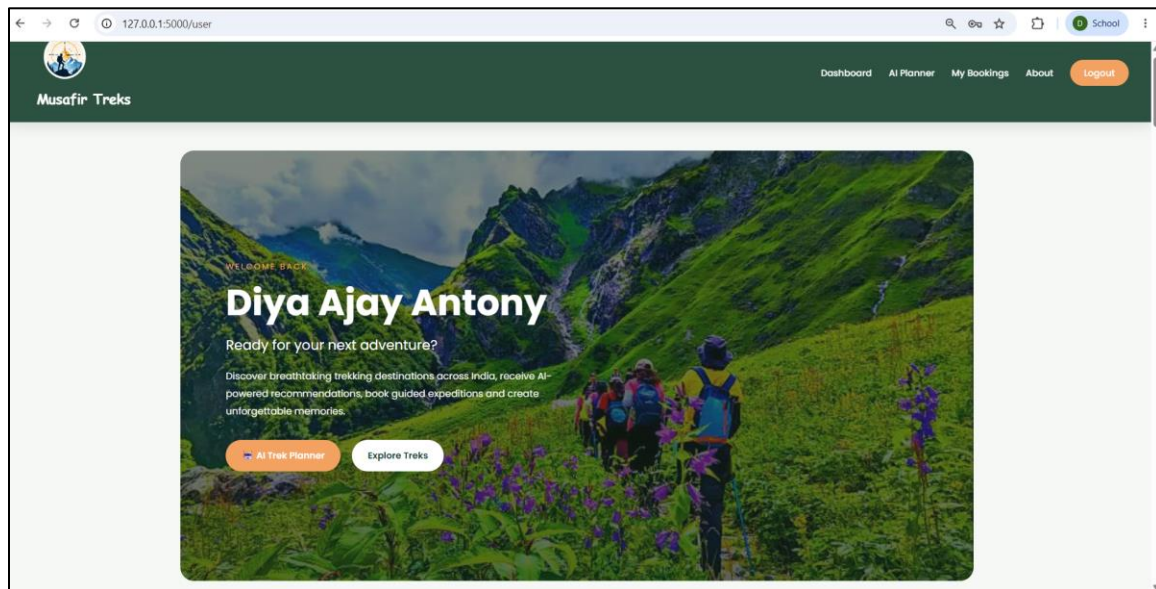


Fig 7.1 User-dashboard

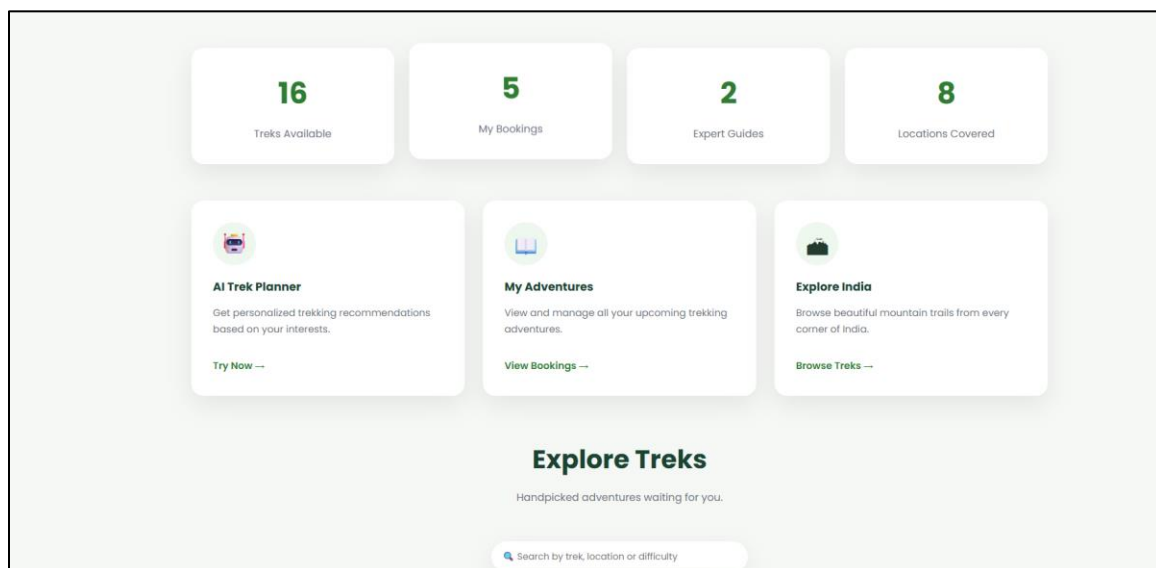


Fig 7.2 User-dashboard-Features

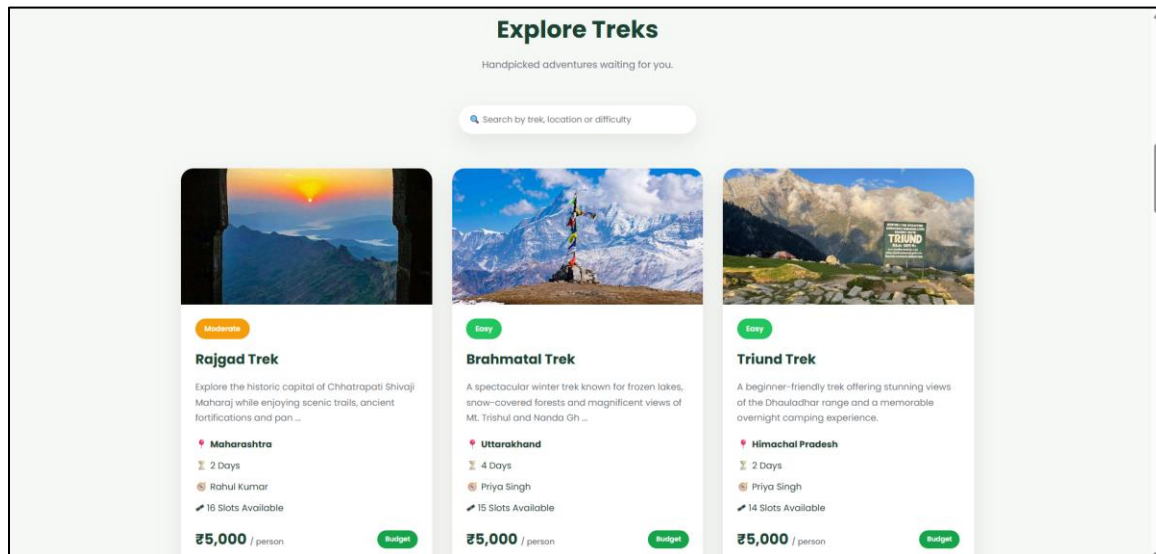


Fig 7.3 Treks Featured

Administrator Module

- Dashboard analytics
- User management
- Staff management
- Trek CRUD operations
- Staff assignment
- Booking monitoring
- Interactive charts

Screenshots of Admin Dashboard:

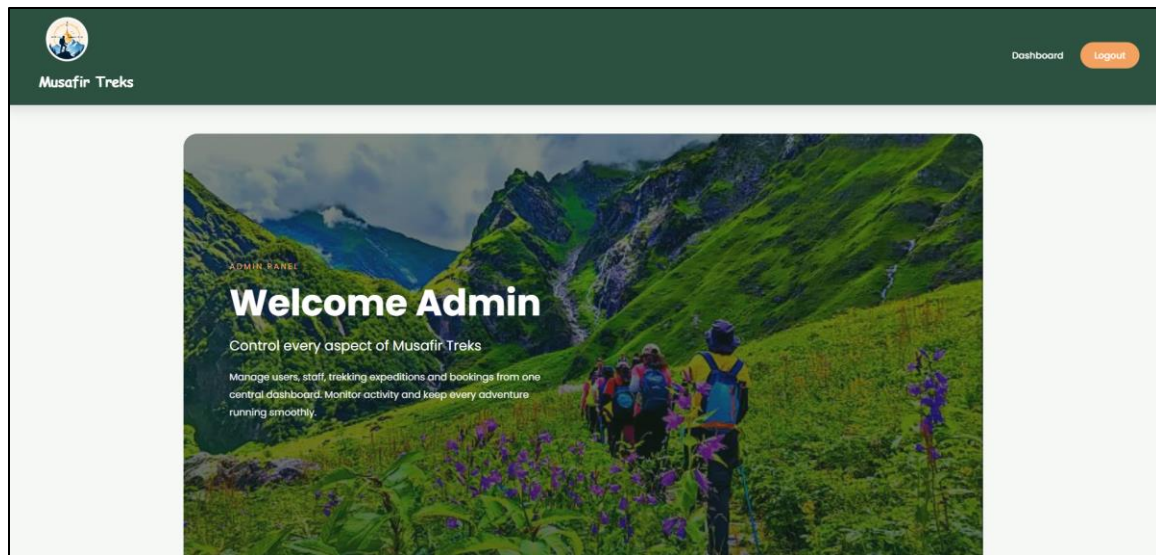


Fig 7.4 Admin-Dashboard

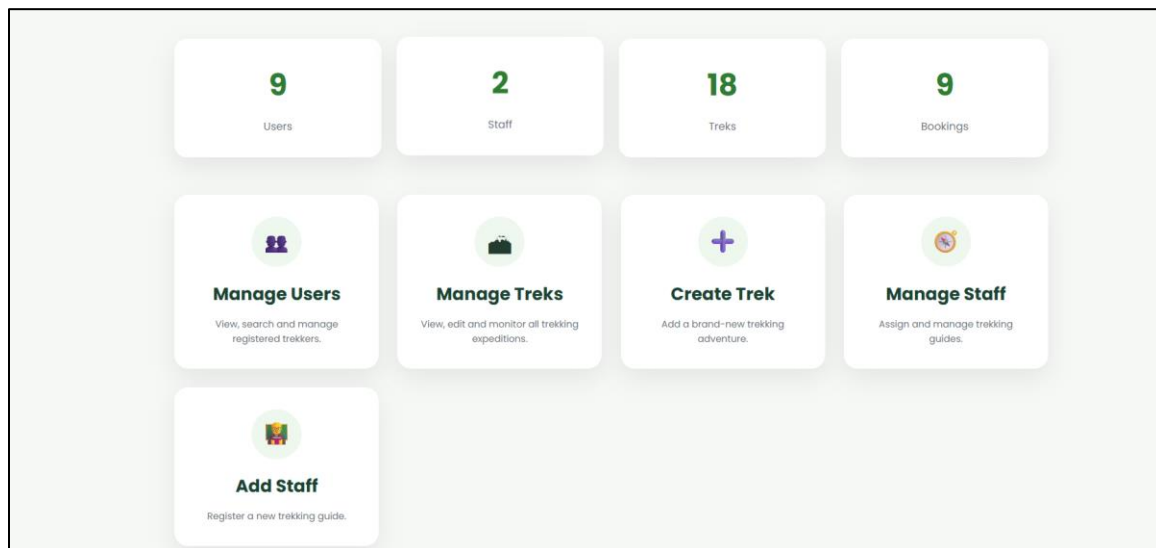


Fig 7.5 Admin-Features

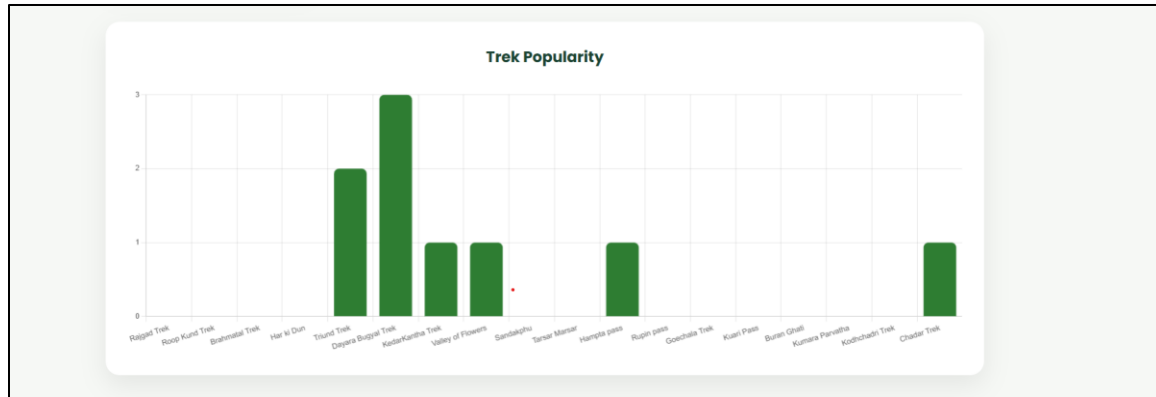


Fig 7.6 Admin-stats

Trek Staff Module

- Dashboard for assigned treks
- Participant management
- Trek status updates
- Occupancy analytics
- Guide reassignment requests

Screenshots of Staff Dashboard:

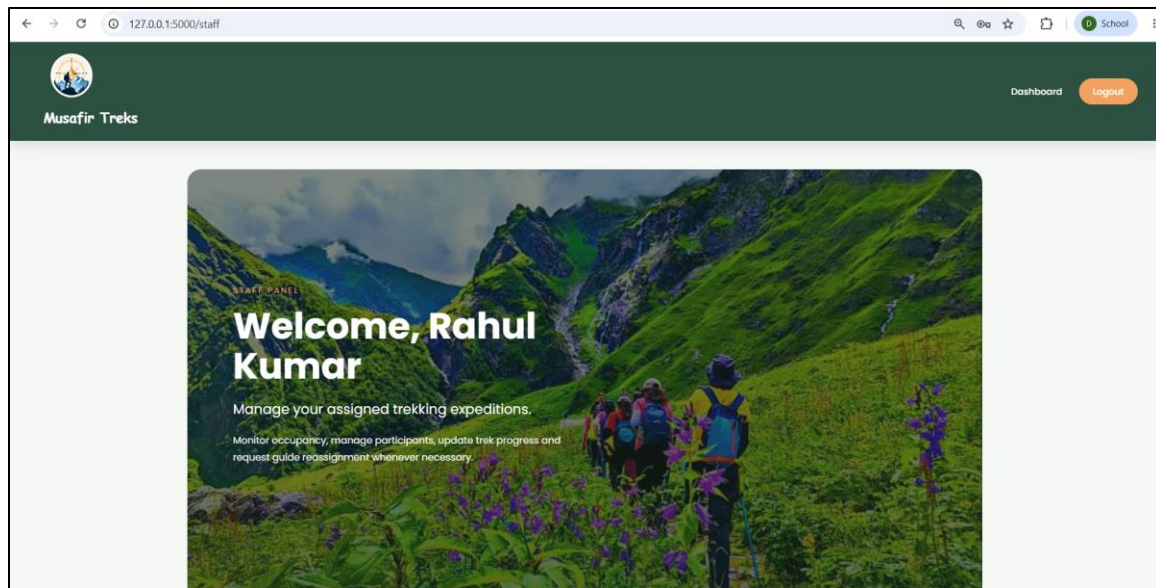


Fig 7.7 Staff Dashboard

7 Assigned Treks								
Trek	Booked	Remaining	Occupancy	Status	Participants	Guide	Progress	
 Rajgad Trek Maharashtra Moderate	0	16	0%	Open	Participants	Request Change	Start Trek	
 Har ki Dun Uttarakhand Easy	0	7	0%	Completed	Participants	Request Change	Completed	
 Sandakphu West Bengal Moderate	0	20	0%	Open	Participants	Request Change	Start Trek	
 Hampta pass Himachal Pradesh Moderate	1	9	10%	Open	Participants	Request Change	Start Trek	
 Goechala Trek Sikkim Difficult	0	8	0%	Open	Participants	Request Change	Start Trek	

Fig 7.8 Staff operations panel and trek Details

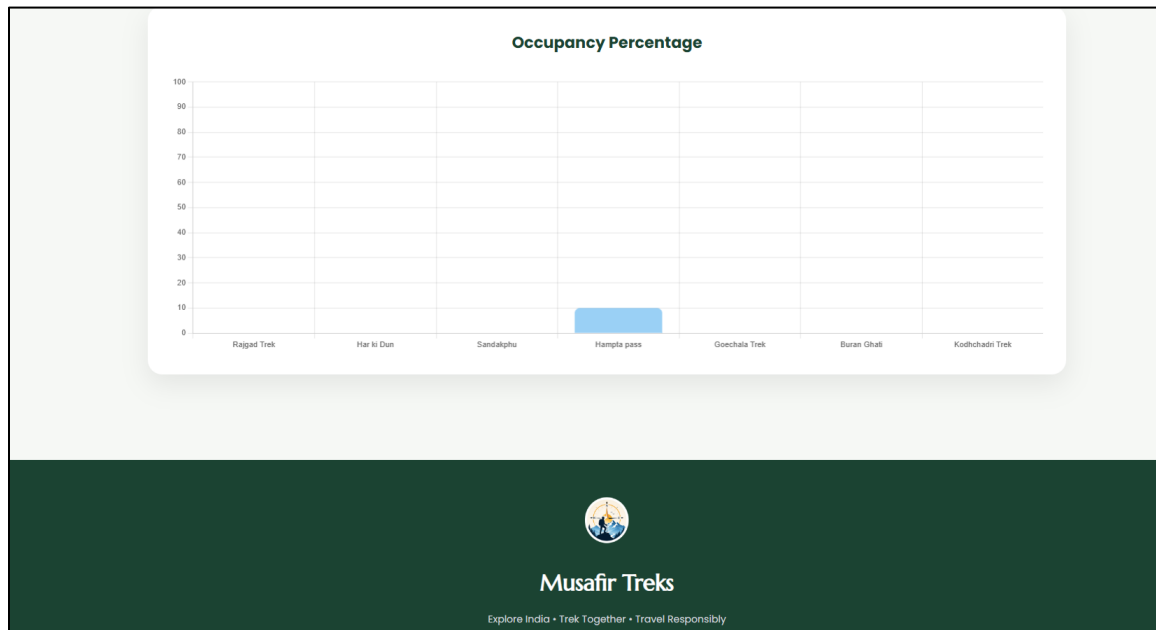
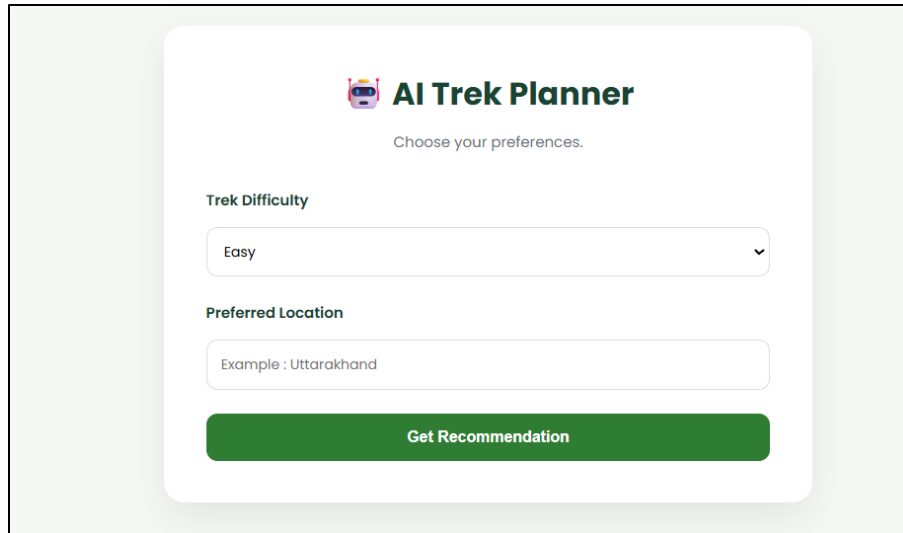


Fig 7.9 Staff-assigned-treks occupancy

Additional Features

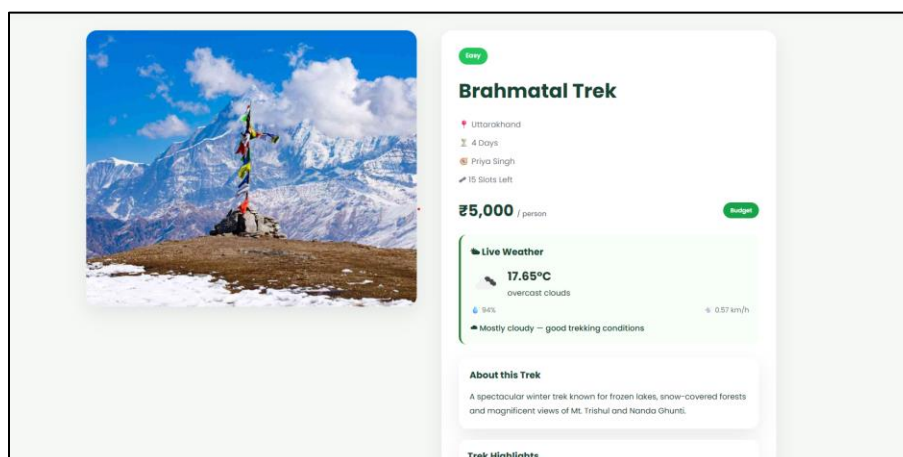
- Live Weather API Integration
- AI-assisted Trek Recommendation
- Budget-friendly Trek Pricing
- Dynamic Trek Descriptions
- Interactive Analytics using Chart.js
- Responsive User Interface
- Custom Musafir Branding

Screenshots of some Features:



The image shows a web interface for an "AI Trek Planner". At the top, there is a small robot icon followed by the title "AI Trek Planner". Below the title, it says "Choose your preferences." There are two input fields: "Trek Difficulty" with a dropdown menu currently set to "Easy", and "Preferred Location" with a text input field containing "Example : Uttarakhand". At the bottom, there is a green button labeled "Get Recommendation".

Fig 7.10 AI trek planner/ recommender



The image displays a detailed view of a trek recommendation for "Brahmatal Trek". On the left is a photograph of a mountain peak with a prayer flag. To the right, the trek details are listed: "Uttarakhand", "4 Days", "Priya Singh", and "15 Slots Left". The cost is shown as "₹5,000 / person" with a green "Budget" tag. Below this is a "Live Weather" section showing "17.65°C", "overcast clouds", "94% humidity", and "0.57 km/h wind". A note states "Mostly cloudy — good trekking conditions". An "About this Trek" section describes it as a "spectacular winter trek known for frozen lakes, snow-covered forests and magnificent views of Mt. Trishul and Nanda Ghunti". At the bottom, there is a "Trek highlights" section.

Fig 7.11 Live Weather, features and costs

8.Challenges faced

During the development of Musafir Treks, several technical challenges were encountered and successfully resolved:

- Designing a scalable role-based authentication system for three different user roles.
- Implementing secure password hashing and session management.
- Preventing duplicate bookings and overbooking.
- Managing database relationships using SQLAlchemy ORM.
- Integrating live weather information using the OpenWeather API.
- Developing AI-assisted trek recommendations based on user preferences.
- Designing responsive dashboards for administrators, trek staff, and users.

9. Future enhancements

The application can be further extended by incorporating:

- Online payment gateway integration.
- Interactive maps using Google Maps API.
- Email and SMS notifications.
- QR-code based trek check-in.
- Personalized AI itinerary planning.
- Mobile application support.

- Trek review and rating system.

10.Video Presentation

Google Drive Link:

<https://drive.google.com/file/d/1VXuJfj0idBoKYpW2a6j4Y8OOShXwF2Sa/view?usp=sharing>

Conclusion

Musafir Treks successfully fulfills the core objectives of a trekking management platform by integrating secure authentication, efficient booking management, administrative analytics, AI-assisted recommendations, and real-time weather information into a unified web application. The project demonstrates the practical application of full-stack web development concepts using Flask, SQLAlchemy, SQLite, and modern frontend technologies while providing a scalable foundation for future enhancements.